

Shivane Dadi

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EDUCATION

University of California, Santa Barbara

M.S. in Computer Science, emphasis in Quantum Computing

B.S. in Computer Science, Dean's Honors

Santa Barbara, CA

Expected June 2027

June 2026

- GPA : 4.00 (M.S., in progress) | 3.66 (B.S.), 3.82 upper-division Computer Science
- ADVISOR : Prof. Murphy Yuezhen Niu, Quantum Codesign Lab
- GRADUATE COURSEWORK : Tensor Computation, Quantum Control, Post-Quantum Cryptography
- UNDERGRADUATE COURSEWORK : Quantum Mechanics I–III, Quantum Computing & Algorithms, Linear Algebra, Group Theory, Probability and Statistics, Data Structures & Algorithms, Numerical Analysis

RESEARCH & APPLIED PROJECTS

Feedforward Error Mitigation for Quantum Simulation

UCSB Quantum Codesign Lab, advised by Prof. Murphy Niu

February 2026 – Present

Santa Barbara, CA

- Developed an algorithmic-level scheme for correcting **time-dependent** control errors in Quantum Signal Processing, errors that are prevalent in long-duration experiments used for quantum simulation.
- Designed a **feedforward drift corrector** by repurposing the circuit's own ancilla measurements as an in-situ sensor of control errors, feeding a sliding-window least-squares rate estimate forward as a predictive phase correction with lightweight periodic diagnostics.
- Proved the scheme runs at $O(1)$ measurement overhead independent of system size.
- Recovered ground-state fidelity from 0.68 to 0.98 under realistic superconducting drift rates; a cost model with zero fitted parameters reproduced the observed overhead across 12 configurations.
- Presented at UCSB CS end-of-year research showcase. Manuscript in preparation.

Noise Robustness of Phase-Factor Finding in QSP

UCSB Quantum Codesign Lab, advised by Prof. Murphy Niu

April 2025 – January 2026

Santa Barbara, CA

- Developed a first-order analytic framework for how hardware noise propagates through **Quantum Signal Processing**, decomposing four realistic error channels as unitary perturbations of the phase factors and deriving a distinct controlling quantity for the resulting error in each regime.
- Applied Fourier and Chebyshev analysis to the QSP phase-factor sequence to show that some error channels concentrate in a small number of circuit layers, rather than uniformly across the circuit.
- Computed the root structure of competing phase-factor solutions in the complex plane and compared their magnitude profiles across solvers, establishing a structural criterion for which of the many valid solutions to a target polynomial resist noise.
- Presented at KITP conference on quantum advantage in the pre-fault-tolerant era, 2026.

Tensor-Network Simulation of Non-Markovian Dynamics

Course research project, ECE 273 (Tensor Computation), UC Santa Barbara

September 2025 – December 2025

Santa Barbara, CA

- Developed a tensor network framework for simulating quantum circuits under **non-Markovian (memory-correlated)** noise, treating gates as interventions injected into continuous-time open system evolution.
- Derived process tensors from a physical spin–boson model in which bath memory correlates every pair of time steps, making exact propagation exponential in circuit depth; applied the **TEMPO** algorithm to compress this multi-time object into a matrix product operator with tunable bond dimension.
- Injected gates as **Liouville-space superoperators** contracted into individual process-tensor time slices, preserving CPTP structure and amortizing the bath computation across arbitrary gate sequences; validated on Ramsey interferometry against a Bloch–Redfield Lindblad baseline.

OpenTGPTPU

UCSB ArchLab, advised by Prof. Jonathan Balkind

August 2024 – June 2025

Santa Barbara, CA

- Implemented **compiler-level parallelism** for memory-disjoint tensor operations on a general-purpose tensor processing unit, enabling safe concurrent scheduling and **reducing compile time by 26%** on large-scale benchmarks.

- Extended the compiler’s intermediate representation and runtime scheduling logic to support parallel op traversal without introducing race conditions or memory contention.
- Designed and built a **custom fuzzer** for regression testing of compiler optimizations, identifying edge cases and faulty op combinations across hundreds of randomized test cases.

Analysis of Cryptographic Verification Protocols

March 2025 – June 2025

Course research report, CMPSC 292G (Post-Quantum Cryptography), UC Santa Barbara

Santa Barbara, CA

- Analyzed candidate protocols for classical of quantum systems with emphasis on efficiency, correctness, and security guarantees
- Studied adversarial models and implemented attacks against proposed protocol variants
- Evaluated tradeoffs between computational cost (both classical and quantum) and verification strength

TEACHING EXPERIENCE

Undergraduate Teaching Assistant

UC Santa Barbara

Department of Computer Science

March 2026 – June 2026

- Automata and Formal Languages (CMPSC 138) with Prof. Prabhanjan Ananth, ~65 students
- Developed homework and exam problems in collaboration with the course instructor
- Assisted students with homework and exam preparation during weekly office hours
- Led discussion sections and graded written homework, designing rubrics to keep grading consistent across the course staff

TALKS AND POSTERS

- **Graduate Simulation Seminar Series (GS³)**: Presented on drift correction in quantum ground-state preparation to a cross-disciplinary audience of graduate students and postdocs; Center for Scientific Computing, UC Santa Barbara, Summer 2026
- **KITP Conference Poster**: *Beyond Gate-Based Quantum Computing: Defining Quantum Advantage in the Pre-Fault Tolerant Era*, Kavli Institute for Theoretical Physics, UC Santa Barbara, May 2026 (with D. Patel and G. Castelazo)
- **UCSB CS Research Showcase**: Presented work on noise robustness in Quantum Signal Processing, UC Santa Barbara, 2026

INDUSTRY EXPERIENCE

Panasonic Avionics

Irvine, CA

Machine Learning Engineering Intern

Summer '24, Summer '25

- Built an agentic **GraphRAG** retrieval system over 300+ internal technical documents (**Neo4j**, **LangChain**), deployed company-wide as a developer support tool.
- Designed an evaluation benchmark from internal support tickets and ran a controlled comparison against the existing vector-only retrieval system, measuring a **17% improvement** in precision and recall.
- Instrumented the pipeline in **MLflow** to track retrieval and generation quality as separate metrics, isolating which stage drove end-to-end performance changes.

SERVICE AND LEADERSHIP

- **Co-founder**, UCSB ACM student chapter 2023-2025: Largest community for ~ 600 CS students at UCSB.
- **Resident Assistant**, UC Santa Barbara Housing 2023-2026

TECHNICAL SKILLS

Core Areas: Quantum algorithms (QSP, QETU, QSVT), Quantum Simulation, Scientific Computation, Open Quantum Systems

Mathematics: Linear algebra, Fisher information and estimation theory, perturbation theory, asymptotic analysis, group theory

Programming: Python, C/C++, Bash, Git, Linux, L^AT_EX

Libraries: NumPy, SciPy, Qiskit, Cirq, PennyLane, OQuPy, PyTorch, Matplotlib